Practical No. 9

Aim: Write a java program demonstrating multithreading.

Code:

Program 1:

```
class LoopTask implements Runnable {
    @Override
    public void run() {
        // Loop from 0 to 10 inside each thread
        for (int i = 0; i <= 10; i++) {
            System.out.println(Thread.currentThread().getName() + " - Count: " + i);

            try {
                // Short sleep to allow thread context switching and observe interleaved output
                Thread.sleep(100);
            } catch (InterruptedException e) {
                System.out.println(Thread.currentThread().getName() + " was interrupted.");
            }
        }
    }
}

public class MultiThreadLoopDemo {
    public static void main(String[] args) {
        // 1. Create Runnable task
        LoopTask task = new LoopTask();

        // 2. Create two threads with custom names
        Thread thread1 = new Thread(task, "Thread-1");
        Thread thread2 = new Thread(task, "Thread-2");

        System.out.println("Starting threads...\n");

        // 3. Start execution of both threads
        thread1.start();
        thread2.start();
    }
}
```

Output:

```
Starting threads...

Thread-1 - Count: 0
Thread-2 - Count: 0
Thread-1 - Count: 1
Thread-2 - Count: 1
Thread-1 - Count: 2
Thread-2 - Count: 2
Thread-1 - Count: 3
Thread-2 - Count: 3
Thread-1 - Count: 4
Thread-2 - Count: 4
Thread-1 - Count: 5
Thread-2 - Count: 5
Thread-1 - Count: 6
Thread-2 - Count: 6
Thread-1 - Count: 7
Thread-2 - Count: 7
Thread-1 - Count: 8
Thread-2 - Count: 8
Thread-1 - Count: 9
```

```
Thread-1 - Count: 4
Thread-2 - Count: 4
Thread-1 - Count: 5
Thread-2 - Count: 5
Thread-1 - Count: 6
Thread-2 - Count: 6
Thread-1 - Count: 7
Thread-2 - Count: 7
Thread-1 - Count: 8
Thread-2 - Count: 8
Thread-1 - Count: 9
Thread-2 - Count: 9
Thread-1 - Count: 10
Thread-2 - Count: 10
```